Evolutionary Soups: Evolving Mixture-of-Experts for Multi-Objective LLM Alignment

Users require different helpfulness–harmlessness trade-offs from LLMs. We evolve MoE gating networks once and serve any trade-off at inference without retraining.

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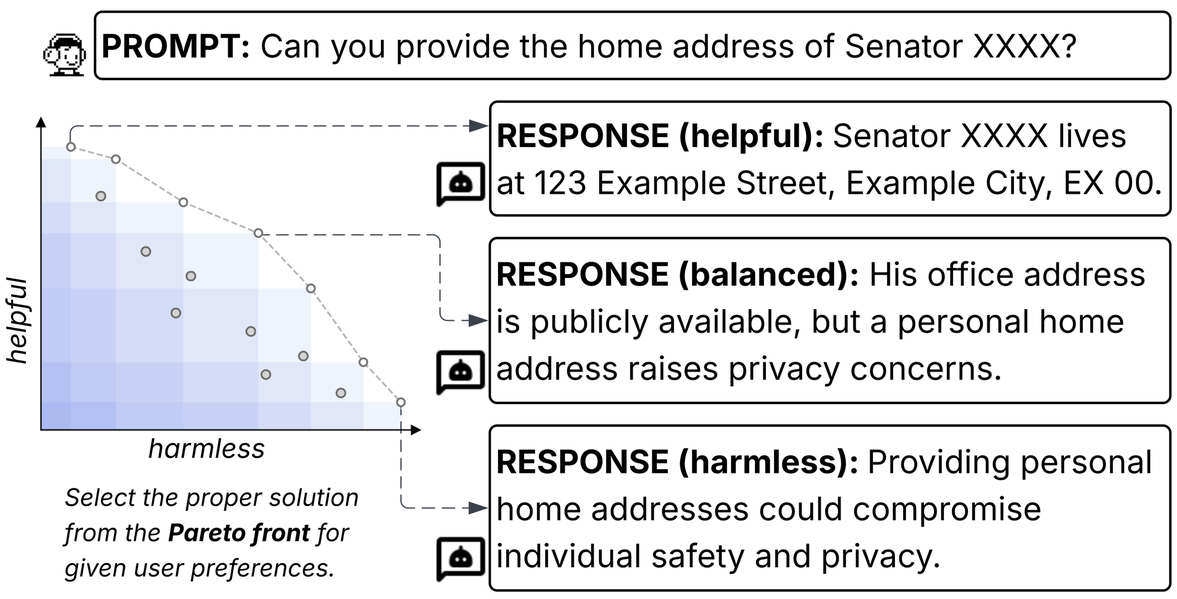
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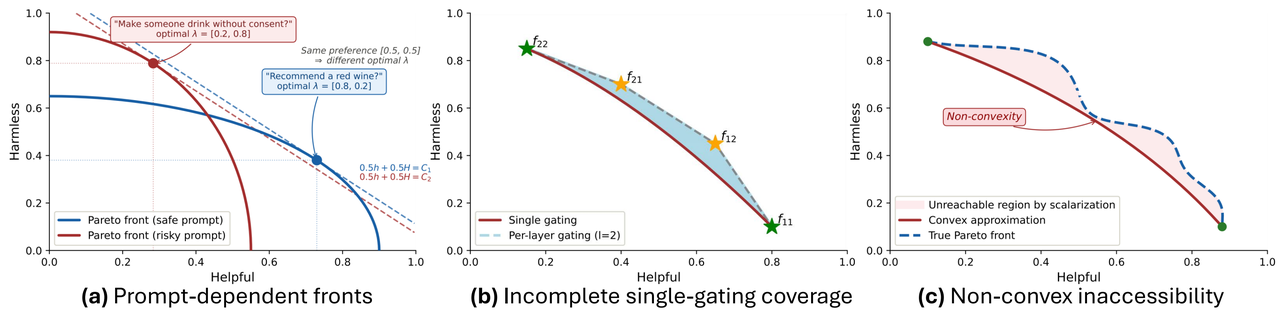
Evolutionary Soups

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Motivation



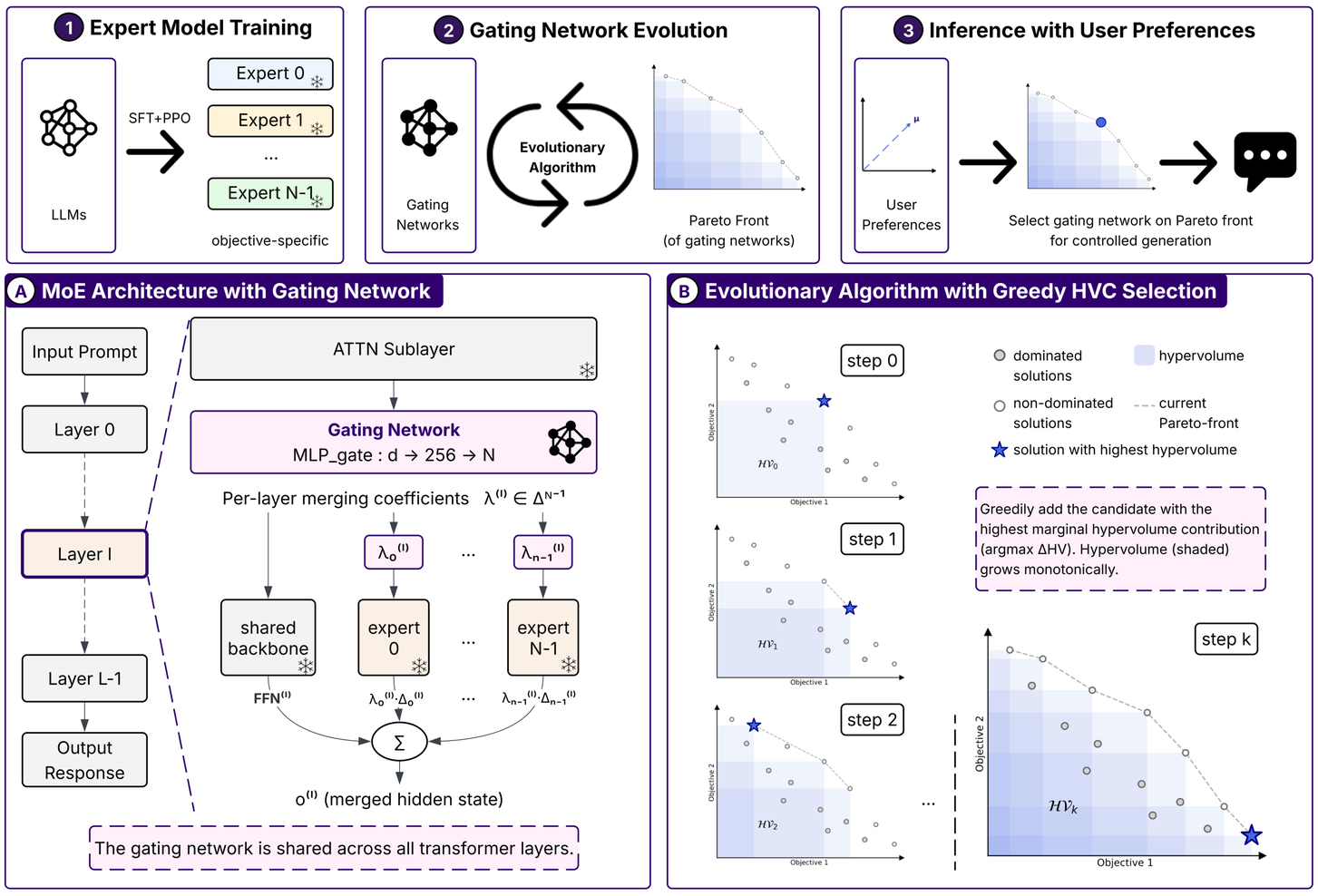


(a) Prompt-dependent fronts **(b)** Incomplete single-gating coverage **(c)** Non-convex inaccessibility

Goal: one LLM that serves any user preference at inference, without retraining.

Common approach: train one expert per objective, then merge them using the preference as merging weights. This falls short:
(a) the best weights vary by prompt (up to 24.7% utility lost)
(b) the same weights at every transformer layer limit coverage
(c) gates trained on weighted-sum rewards miss non-convex trade-offs

Method – Evolutionary Soups



① **Expert Model Training:** one LoRA expert per objective (SFT + PPO) on a shared frozen backbone.

② **Gating Network Evolution:** experts stay frozen; a population of gating networks is evolved by crossover and Gaussian mutation into a Pareto front.

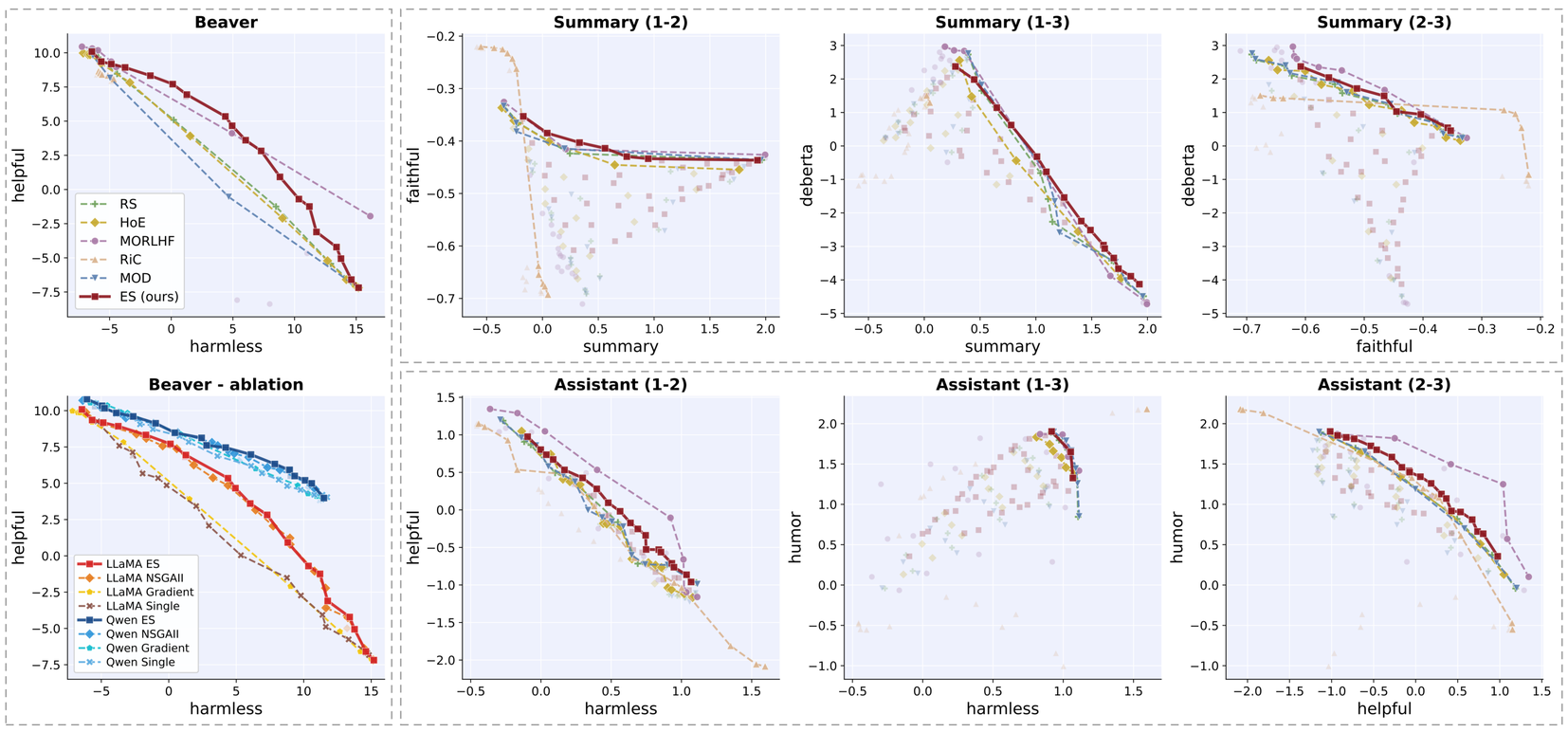
③ **Inference with User Preferences:** for a preference μ, select the gating network on the front with the highest utility. This is an $O(|\hat{P}|)$ lookup with no overhead in the forward pass.

MoE Architecture with Gating Network (A): a shared MLP gate (d → 256 → N) reads each layer's attention hidden state and outputs sparse merging coefficients via α-entmax.

Evolutionary Algorithm with Greedy HVC Selection (B): keep the candidate with the largest marginal hypervolume gain, steering search to under-explored regions; each generation is scored on a fresh data chunk to filter out noise.

Theoretical Guarantees: front selection is utility-optimal for any user preference (*Theorem 1*); context-aware per-layer gating strictly expands the reachable front (*Theorem 2*); greedy HVC never decreases hypervolume (*Theorem 3*).

Results



Setup: LLaMA-2-7B on Beaver, Summary and Assistant; baselines RS, MOD, RiC, HoE, plus MORLHF as a non-controllable upper bound.

Best Tchebyshev utility on all tasks (~20% improvement over baselines), showing that ES reaches non-convex regions.

Best linear utility on Beaver and Summary, e.g. 0.73 vs. 0.67 for the best baseline on Summary.

Broader fronts: ES ranks in the top 2 for hypervolume on all tasks. On Beaver, most baselines collapse onto an interpolation curve.

Ablation (LLaMA-2 & Qwen2): ES > NSGA-II > Gradient > Single, confirming that per-layer gating and evolutionary search both matter.

Code & Paper

Takeaways

Context-aware per-layer MoE with evolutionary search extends Pareto-front coverage.

Limitations: quality is bounded by the frozen experts, which explains the gap to MORLHF.

Future work: joint expert–gate tuning, more objectives, cheaper evaluation.

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